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Roll No

EC-603 (GS)

B.E. VI Semester Examination, June 2020

Grading System (GS)

Digital Signal Processing

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Explain and differentiate the following :
 - i) Causal versus noncausal system
 - ii) Linear versus nonlinear system

2. Determine $H(z)$ and its poles and zeros if

$$y(n) + \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n) + x(n-1)$$

3. Determine the cross correlation sequence $r_{xy}(l)$ of the sequences

$$x(n) = \left\{ \cdots 0, 0, 2, \underset{\uparrow}{-1}, 3, 2, 1, 2, -3, 0, 0, \cdots \right\}$$

$$y(n) = \left\{ \cdots 0, 0, 1, -1, 2, -2, 4, 1, -2, 5, 0, 0, \cdots \right\}$$

4. Write short notes on any two of the following:
 - a) FFT algorithm
 - b) Windowing technique
 - c) Frequency domain representation of signal and systems.
5. Explain the procedure for designing a FIR filter using the Kaiser window.
6. What is impulse invariant technique? Obtain the mapping formula for the impulse invariant transformation.
7. Draw and explain the graphical representation of Discrete Time Signal.
OR
Explain the difference between decimation in time and decimation in frequency FFT algorithm.
8. Derive the Z-transform of $f(nT) = \sin \omega nT$.
